

Notes: Liu, Makridis, Ouimet & Simintzi (RFS, 2023)

The Distribution of Nonwage Benefits: Maternity Benefits and Gender Diversity

Contents

1	Big picture	3
2	Data and measurement	3
2.1	Glassdoor nonwage benefit reviews	3
2.2	Baseline public-firm maternity sample	4
2.3	Maternity benefit rating	4
2.4	Female high-skill labor supply	4
2.5	Firm controls	5
2.6	Female-friendly and gender-neutral benefits	5
2.7	Paid weeks of maternity leave	5
2.8	Female employment and gender composition	5
2.9	State paid family leave laws and stock returns	6
3	Models and empirical specifications	6
3.1	Sorting model intuition	6
3.2	Baseline maternity benefit regression	6
3.3	Robustness specifications	7
3.4	Female-friendly versus gender-neutral benefit regressions	7
3.5	Paid weeks specification	7
3.6	State-level regressions	7
3.7	Instrumental-variable specification	8
3.8	Female employment mechanism	8
3.9	Event-study specification for firm value	8
4	Identification and empirical design	8
4.1	Main identifying variation	8
4.2	Why maternity leave is a clean benefit to study	9
4.3	Ruling out general firm generosity	9
4.4	External validity strategies	9
4.5	Firm-value identification	9
4.6	Limitations	10
5	Tables and figures	11
5.1	Table 1 and Figures 1–2: benefit ratings across categories, time, and industries	11
5.2	Table 2: baseline sample summary statistics	12
5.3	Table 3: baseline industry female high-skill result	12
5.4	Table 4: robustness tests	13
5.5	Table 5: female-friendly versus gender-neutral benefits	14
5.6	Table 6: actual paid weeks of maternity leave	15
5.7	Table 7 and Table 8: state-level and IV evidence	16
5.8	Table 9: maternity benefits and female employment	16
5.9	Table 10: event study around paid family leave laws	17

5.10	Appendix tables: sample construction and variable definitions	17
6	Short summary	18
7	Conclusion	19
7.1	Core conclusion	19
7.2	Economic intuition	19
7.3	Takeaway	19

1 Big picture

- **Question.** Why do firms offer costly nonwage benefits instead of simply paying equivalent cash compensation? The paper studies whether firms use female-friendly benefits, especially maternity leave, to attract and retain scarce female talent.
- **Core hypothesis.** Firms operating in labor markets with relatively low supply of high-skilled women offer more generous maternity benefits because those benefits are especially valuable to female employees and can help firms increase gender diversity.
- **Economic mechanism.** Nonwage benefits can be a screening and sorting device. A benefit that is more valuable to one type of worker than another can help a firm attract the desired type without paying unequal wages or relying only on cash compensation.
- **Main data innovation.** The paper uses Glassdoor benefit reviews, which include benefit-specific ratings, respondent characteristics, firm identifiers, and location information. This allows the authors to study benefit quality rather than only whether a benefit exists.
- **Main baseline result.** Maternity benefit ratings are lower in industries with a larger share of high-skilled female workers. Equivalently, firms provide more generous maternity benefits when high-skilled female talent is scarcer.
- **Economic magnitude.** A one-standard-deviation increase in the female share of college-educated workers in the industry is associated with about a 0.09–0.10 point reduction in maternity benefit ratings, roughly 0.07–0.08 standard deviations of the rating variable.
- **Generalization.** The same logic holds for a broader set of female-friendly benefits, but not for gender-neutral benefits. This supports the sorting interpretation rather than a general firm generosity story.
- **External validity.** Results are robust across industry-level and state-level labor markets, with instrumental-variable evidence using WWII mobilization rates to predict state-level female high-skill labor supply.
- **Mechanism evidence.** Firms with better maternity benefits have higher female employment in industries where female talent is scarce. This links benefits to gender diversity rather than just to survey ratings.
- **Firm value evidence.** State paid family leave mandates reduce the relative value of voluntary maternity benefits for firms that had already used these benefits to compete for scarce female talent. Treated high-maternity firms in low female-skill labor markets experience negative abnormal returns around law passage.
- **Takeaway.** Nonwage benefits are not merely perks. Firms can use them strategically to target and retain specific types of workers, and the value of these benefits depends on labor-market scarcity.

2 Data and measurement

2.1 Glassdoor nonwage benefit reviews

- The core data come from Glassdoor employer benefit reviews from July 2014 through April 2019.
- Employees assign ratings from 1 to 5 for specific benefit categories.
- The data include maternity leave ratings and many other nonwage benefit categories.
- Respondents report demographic and job-related characteristics, including gender, age, education, base pay, and whether the reviewed job is current or past.
- The baseline public-firm sample is restricted to regular employees, excluding part-time, contractor, and other nonregular workers for whom benefits may not be available.

Interpretation: Glassdoor data allow the authors to observe employee-perceived benefit quality at the benefit level. This is richer than standard compensation data, which often record wages and benefits expenditure but not perceived quality or employee type.

2.2 Baseline public-firm maternity sample

The main baseline sample is constructed as:

Glassdoor maternity reviews $\rightarrow$ regular employees $\rightarrow$ nonmissing respondent data $\rightarrow$ public firms matched to Compustat. (1)

- The raw Glassdoor maternity benefit review data include 45,217 observations.
- Dropping part-time, contract, and other nonregular employees leaves 40,696 reviews.
- Requiring nonmissing respondent characteristics leaves 10,385 maternity reviews.
- Restricting to public firms matched to Compustat leaves 3,157 observations.
- The public-firm sample represents 1,094 unique publicly listed firms.
- The average firm has 2.2 maternity benefit reviews per firm-year.

Interpretation: The baseline sample is smaller than the full Glassdoor sample but permits controls for firm assets, profitability, leverage, and Tobin’s q . This improves identification at the cost of external validity to private and small firms.

2.3 Maternity benefit rating

The main dependent variable is

$$MaternityRating_{i,f,t} \in \{1, 2, 3, 4, 5\}, \quad (2)$$

where respondent i rates the maternity benefit quality of firm f in year t .

- The average maternity benefit rating is 3.86.
- The median rating is 4.
- Respondents who rate maternity leave are disproportionately high-skill: 89% have a college degree.
- 46% of maternity benefit reviewers are female.
- 68% review a current job, and 32% review a previous job.

Interpretation: The sample is tilted toward educated, younger, higher-paid workers. This is useful because the theory concerns competition for skilled female talent, but the authors still need robustness checks for sample selection.

2.4 Female high-skill labor supply

The key explanatory variable is the female share of college-educated workers in the relevant labor market:

$$FemaleHighSkillShare_{j,t} = \frac{FemaleCollegeWorkers_{j,t}}{CollegeWorkers_{j,t}}. \quad (3)$$

In the baseline analysis, j is the firm’s four-digit NAICS industry. In later tests, j is a state.

- The average industry female high-skill share is 0.41.

- The 25th percentile is 0.30 and the 75th percentile is 0.52.
- The measure comes from Quarterly Workforce Indicators.

Interpretation: A low value means high-skilled women are scarce in the relevant labor market. The paper predicts that firms should offer better female-friendly benefits in these markets.

2.5 Firm controls

The baseline public-firm controls are:

- $\ln(Assets)$;
- profit margin;
- debt/assets;
- Tobin’s q ;
- in some specifications, respondent base pay and firm gender measures.

Interpretation: These controls address the concern that benefit quality is driven by general firm resources, profitability, financial constraints, or investment opportunities rather than by labor-market scarcity.

2.6 Female-friendly and gender-neutral benefits

The paper classifies benefits into:

- **Female-friendly benefits:** adoption assistance, childcare, dependent care, diversity programs, family medical leave, fertility assistance, flexible accounts, health insurance, mental health care, maternity leave, reduced or flexible hours, vacation, and work from home.
- **Gender-neutral benefits:** benefits such as dental insurance, disability insurance, employee discount, stock purchase plans, retirement plans, stock options, tuition assistance, legal assistance, company car, and similar benefits.

Interpretation: This classification allows a falsification test. If scarce female labor leads firms to target women, the effect should appear for female-friendly benefits, not for all benefits equally.

2.7 Paid weeks of maternity leave

A separate external data source, Fairygodboss, provides actual paid maternity leave weeks:

$$PaidWeeks_f. \tag{4}$$

The paper estimates regressions using $\ln(1 + PaidWeeks_f)$ as the dependent variable.

Interpretation: Glassdoor ratings are subjective; paid weeks provide a more objective measure of maternity benefit generosity.

2.8 Female employment and gender composition

For mechanism tests, the paper uses firms that disclose gender composition through diversity reports or CSR disclosures.

- The dependent variable is the share of female employees at the firm.
- The sample contains 343 firms in the baseline female-employment mechanism test.

Interpretation: This test asks whether generous maternity benefits are associated with actual increases in female representation, especially where female talent is scarce.

2.9 State paid family leave laws and stock returns

The firm-value section studies the passage or expansion of paid family leave laws in states including New York, Washington, Massachusetts, Connecticut, Oregon, Washington DC, California, and New Jersey.

The event-study dependent variable is

$$CAR_{f,e}[-1, +1], \quad (5)$$

where abnormal returns are computed around the law passage event using CAPM or Carhart four-factor expected returns.

Interpretation: The event-study tests whether voluntary maternity benefits had value. If mandates reduce the strategic differentiation value of voluntary benefits, high-maternity firms in low-female-talent markets should lose value when state benefits expand.

3 Models and empirical specifications

3.1 Sorting model intuition

The core mechanism can be written as:

$$FemaleTalentScarcity \Rightarrow HigherMarginalValueOfFemaleWorkers \Rightarrow BetterFemaleFriendlyBenefits. \quad (6)$$

Firms choose nonwage benefits when those benefits are more valuable to target workers than to others:

$$BenefitValue_{female} > BenefitValue_{male}. \quad (7)$$

Interpretation: A female-friendly benefit can be more cost-effective than a wage increase because it is differentially valued by the workers the firm wants to attract.

3.2 Baseline maternity benefit regression

The baseline specification is:

$$Rating_{i,f,t} = \alpha + \beta FemaleHighSkillShare_{j,t} + X'_{i,t}\gamma + Z'_{f,t}\delta + au_t + \eta_s + \varepsilon_{i,f,t}. \quad (8)$$

- $Rating_{i,f,t}$ is the maternity benefit rating.
- $FemaleHighSkillShare_{j,t}$ is measured at the four-digit NAICS industry level.
- $X_{i,t}$ includes respondent characteristics: age, current job, female, college, and sometimes base pay.
- $Z_{f,t}$ includes firm controls: assets, profit margin, debt/assets, and q .
- τ_t are year fixed effects.
- η_s are respondent-state fixed effects in some specifications.

Interpretation: The predicted sign is $\beta < 0$. A lower female high-skill share means women are scarce, so firms should provide higher maternity benefits.

3.3 Robustness specifications

The paper repeats the baseline regression while:

- controlling for the firm share of female executives;
- controlling for the firm gender pay gap;
- keeping only female respondents;
- keeping only past-employee reviews;
- dropping technology and finance firms;
- weighting by the inverse number of firm reviews;
- assigning zero maternity benefit rating to firms reported as not offering maternity benefits.

Interpretation: These tests address concerns about omitted female-friendly culture, pay policy, respondent awareness, Glassdoor review manipulation, industry composition, unequal review intensity, and nonresponse.

3.4 Female-friendly versus gender-neutral benefit regressions

For expanded benefit data, the paper estimates benefit-level regressions of the form:

$$BenefitRating_{i,b,f,t} = \alpha + \beta FemaleHighSkillShare_{j,t} + X'_{i,t}\gamma + Z'_{f,t}\delta + \tau_t + \varepsilon_{i,b,f,t}. \quad (9)$$

It also estimates pooled regressions with an interaction:

$$BenefitRating = \alpha + \beta_1 FemaleHighSkillShare + \beta_2 FemaleFriendly + \beta_3 (FemaleHighSkillShare \times FemaleFriendly) + Controls + \tau_t + \varepsilon_{i,b,f,t}. \quad (10)$$

Interpretation: The central prediction is that female high-skill share should have a stronger negative effect for female-friendly benefits than for gender-neutral benefits.

3.5 Paid weeks specification

For objective paid-leave data, the paper estimates:

$$\ln(1 + PaidWeeks_f) = \alpha + \beta FemaleHighSkillShare_j + Z'_f\delta + \varepsilon_f. \quad (11)$$

Interpretation: A negative coefficient shows that the result is not only a ratings phenomenon. Firms in low-female-talent industries actually offer more paid leave weeks.

3.6 State-level regressions

The paper repeats the baseline idea using state-level female high-skill shares:

$$Rating_{i,f,t} = \alpha + \beta FemaleHighSkillShareState_{s,t} + Controls + \tau_t + \varepsilon_{i,f,t}. \quad (12)$$

Interpretation: State-level tests broaden the definition of labor market and include public and private firms, strengthening external validity.

3.7 Instrumental-variable specification

The state-level IV uses WWII mobilization rates as an instrument for current female high-skill labor supply.

First stage:

$$FemaleHighSkillShareState_s = \pi_0 + \pi_1 WWII Mobilization_s + W'_s \pi + u_s. \quad (13)$$

Second stage:

$$MaternityRating_s = \alpha + \beta FemaleHighSkillShareState_s + W'_s \delta + \epsilon_s. \quad (14)$$

Interpretation: WWII mobilization reduced male labor supply and affected women's human capital decisions. The instrument is used to isolate cross-state variation in female high-skill labor supply.

3.8 Female employment mechanism

The mechanism regression is:

$$FemaleEmployment_f = \alpha + \beta_1 MedianMaternityRating_f + \beta_2 LowFemaleHighSkill_j + \beta_3 (MedianMaternityRating_f \times LowFemaleHighSkill_j) \quad (15)$$

Interpretation: The coefficient of interest is β_3 . The sorting mechanism predicts that better maternity benefits should be associated with higher female employment especially where female talent is scarce.

3.9 Event-study specification for firm value

The event-study specification is:

$$CAR_{f,e} = \beta_1 Law_{s,e} \times HighMaternity_f \times LowFemaleHighSkill_j \quad (16)$$

$$+ \beta_2 Law_{s,e} \times LowFemaleHighSkill_j \quad (17)$$

$$+ \beta_3 HighMaternity_f \times LowFemaleHighSkill_j \quad (18)$$

$$+ \beta_4 Law_{s,e} \times HighMaternity_f + \beta_5 HighMaternity_f + \lambda_{I,e} + \theta_{S,e} + \varepsilon_{f,e}. \quad (19)$$

- $Law_{s,e}$ equals one for firms headquartered in states affected by a paid leave law event.
- $HighMaternity_f$ equals one if pre-treatment maternity benefit ratings are above the relevant median.
- $LowFemaleHighSkill_j$ identifies industries or states where high-skilled female labor is scarce.
- $\lambda_{I,e}$ are industry-event fixed effects.
- $\theta_{S,e}$ are state-event fixed effects.

Interpretation: If voluntary generous maternity benefits created value by differentiating firms in scarce female-talent markets, a public mandate should reduce that differentiation value. The predicted sign is $\beta_1 < 0$.

4 Identification and empirical design

4.1 Main identifying variation

- The key cross-sectional variation is the supply of high-skilled women across industries or states.
- The dependent variable is benefit generosity, either through Glassdoor ratings or paid weeks.
- The main empirical comparison is between firms operating in high versus low female high-skill labor markets.

- The identifying assumption is that after controls, variation in female high-skill labor supply captures scarcity of female talent rather than unrelated firm generosity.

Interpretation: The design is not an RCT. Identification comes from matching the benefit type to the target worker type and showing that results hold only for female-friendly benefits.

4.2 Why maternity leave is a clean benefit to study

- Maternity leave is disproportionately valued by women.
- Firms can voluntarily offer paid maternity benefits because U.S. federal law does not mandate paid maternity leave.
- Maternity benefits can attract women without explicitly paying gender-specific wages.
- Maternity benefits may also signal a female-friendly workplace culture.

Interpretation: These features make maternity leave especially suitable for testing the sorting mechanism.

4.3 Ruling out general firm generosity

- The effect appears for maternity benefits and female-friendly benefits.
- The effect is weaker or absent for gender-neutral benefits.
- Firm controls absorb size, profitability, leverage, and growth opportunities.
- Robustness tests control for female executives and gender pay gap.

Interpretation: If the result were just general generosity, it should appear equally for gender-neutral benefits. The differential pattern supports the targeting interpretation.

4.4 External validity strategies

- Paid weeks data validate that ratings correspond to actual benefits.
- State-level tests broaden the labor-market definition.
- Public+private samples expand beyond Compustat firms.
- IV evidence uses WWII mobilization to address endogeneity in female high-skill supply.

Interpretation: The paper layers several designs to address measurement, sample selection, and endogeneity concerns.

4.5 Firm-value identification

- Paid leave mandates reduce the relative advantage of firms that voluntarily offered generous maternity benefits.
- The paper estimates abnormal returns around law passage events.
- Treated firms are headquartered in states passing or expanding paid family leave laws.
- Control firms are headquartered elsewhere.
- The strongest prediction concerns high-maternity firms in labor markets with scarce high-skilled women.

Interpretation: This is the closest design to a shock. It asks whether the stock market views voluntary maternity benefits as value-relevant when their differentiation value is weakened by public mandates.

4.6 Limitations

- Glassdoor reviewers are not representative of all employees.
- Benefit ratings are subjective and may reflect employee sentiment.
- Female high-skill labor supply could reflect social norms or unobserved local characteristics.
- Firm-level gender composition is available only for firms that disclose diversity information.
- Event-study estimates rely on law-passage timing and state headquarters definitions.

Interpretation: The paper's strength comes from many mutually reinforcing tests rather than from a single clean experiment.

5 Tables and figures

The visuals below are directly extracted from the paper and grouped to keep the notes compact.

5.1 Table 1 and Figures 1–2: benefit ratings across categories, time, and industries

Read:

- Table 1 shows correlations between benefit categories and firm characteristics.
- Firm size tends to be positively related to benefit quality.
- Figure 1 shows average ratings across time for benefit groups.
- Flexibility, retirement, and maternity benefits are among the highest-rated categories.
- Figure 2 shows industry variation in benefit quality.

Economic message: Benefit quality differs across firms, industries, and benefit categories. This heterogeneity motivates the question of why firms choose different nonwage benefit packages.

Table 1
Correlation between benefit ratings and firm characteristics

	ln(Assets)	Profit margin	Debt/Assets	q	ln(Base pay)
Health	0.077	−0.012	−0.142	0.135	0.155
Life and casualty insurance	0.123	0.019	−0.081	0.105	0.151
Retirement	0.195	0.129	−0.024	−0.018	0.038
Equity-based compensation	0.019	0.030	−0.145	0.167	0.108
Training and education	0.090	0.048	−0.120	0.071	0.117
Leave and vacation	0.129	0.029	−0.132	0.033	0.168
Flexibility	−0.049	0.002	−0.057	0.121	0.164
Perks	−0.056	0.012	−0.082	0.109	−0.036
Maternity	0.100	0.001	−0.065	0.085	0.096

This table displays pairwise correlations between benefit group ratings and firm-level characteristics. The sample reflects all employer benefit reviews from July 2014 through April 2019 from respondents with nonmissing individual characteristics of firms that can be matched to Compustat. We require that all respondents in the sample are regular employees of the firm, either currently or previously. We drop part-time, contractor, and other nonregular employees, for whom benefits may not be available. All variables are collapsed to the firm level as averages. Correlations are calculated at the firm level between average benefit group ratings and firm characteristics. Appendix A.2 describes all variables. Standard errors are clustered at the industry level. Shading represents significance at 5% (lighter) and 1% (darker) levels.

observe detailed information for the employees providing the ratings. For example, whereas the publicly available version on its website reports only the mean and range of salaries for a given occupation and firm, we observe granular salary information for individuals in a given occupation and firm. The website also reports information on benefit ratings but does not allow for these ratings to be linked to employee characteristics.

Despite the importance of nonwage compensation, our understanding of the distribution of nonwage benefits at firms is limited. Prompted by this scarcity of evidence, we first present descriptive statistics using broad nonwage benefits categories we created based on our Glassdoor data. For ease of exposition, we have grouped the 54 unique benefit categories from Glassdoor into 8 broad benefit groups: Health, Life and Casualty Insurance, Retirement, Equity-Based Compensation, Training and Education, Leave and Vacation, Flexibility, and Perks. To facilitate comparison with our baseline analysis to follow, we keep maternity benefits—the focus of our paper—as a stand-alone category.

Table 1: correlations between benefit ratings and firm characteristics

Figure 1

Average ratings by benefit groups over time

This figure displays average ratings by benefit group over time. The sample reflects all employer benefit reviews from July 2014 through April 2019 from respondents with nonmissing individual characteristics of firms that can be matched to Compustat. We require that all respondents in the sample are regular employees of the firm, either currently or previously. We drop part-time, contractor, and other nonregular employees for whom benefits may not be available. The dark-blue series represents reviews from 2014 through 2016. The light-blue series represents reviews from 2017 through 2019. Benefit groups are defined as follows. Health: Dental Insurance, Fertility Assistance, Flexible Spending Account (FSA), Health Insurance, Health Savings Account (HSA), Mental Health Care, and Vision Insurance. Life and Casualty Insurance: Disability Insurance, Life Insurance, Occupation Accident Insurance, Supplemental Life Insurance, and Supplemental Workers' Compensation. Retirement: 401K Plan, Pension Plan, Retiree Health & Medical, and Retirement Plan. Equity-Based Compensation: Employee Stock Purchase Plan, Equity Incentive Plan, Performance Bonus, and Stock Options. Training and Education: Apprenticeship Program, Job Training, Professional Development, and Tuition Assistance. Leave and Vacation: Bereavement Leave, Family Medical Leave, Military Leave, Paid Holidays, Sabbatical, Sick Days, Unpaid Extended Leave, Vacation & Paid Time Off, and Volunteer Time Off. Flexibility: Reduced or Flexible Hours and Work from Home. Perks: Adoption Assistance, Charitable Gift Matching, Childcare, Commuter Checks & Assistance, Company Car, Company Social Events, Dependent Care, Diversity Program, Employee Assistance Program, Employee Discount, Free Lunch or Snacks, Gym Membership, Health Care On-Site, Legal Assistance, Mobile Phone Discount, Pet-Friendly Workplace, and Travel Concierge. Maternity: Maternity Leave.

often significant correlations between investment opportunities (q) and benefit quality. Interestingly, the benefit group with the highest correlation with q is Equity-Based Compensation, which includes benefits commonly associated with the high-growth technology sector. Ittner, Lambert, and Larcker (2003) find that these firms place particular importance on employee retention, relative to firms in more established industries, potentially explaining this correlation.

We also find that firm-level average base pay ($\ln(\text{Base pay})$) is positively correlated with benefit quality across the majority of benefit groups. It is intuitive that workers who receive higher cash wages also receive higher noncash benefits, on average. Our subsequent analysis mostly focuses on maternity leave, a specific nonwage benefit, though maternity leave interestingly behaves similarly to other benefit categories when it comes to

Figure 1: benefit ratings over time

Figure 2**Average ratings by industry**

This figure displays average ratings by industry and time. The sample reflects all employer benefit reviews from July 2014 through April 2019 from respondents with nonmissing individual characteristics of firms that can be matched to Compustat. We require that all respondents in the sample are regular employees of the firm, either currently or previously. We drop part-time, contractor, and other nonregular employees, for whom benefits may not be available. The dark-blue series represents reviews from 2014 through 2016. The light-blue series represents reviews from 2017 through 2019. Industries displayed in this figure correspond to one-digit NAICS industry classifications, with the exception of Agriculture, Natural Resources, Utilities, and Construction, which consists of one-digit NAICS=1 and one-digit NAICS=2 because of small sample sizes.

(2017–2019) is higher than the first half (2014–2016). This trend is consistent with the increasing share of total compensation that nonwage compensation composes; greater spending on nonwage compensation over time translates into improved benefit quality. Moreover, we document the largest increase in average maternity benefit ratings.

5.2 Table 2: baseline sample summary statistics

Read:

- The average maternity benefit rating is 3.86 and the median is 4.
- 46% of reviewers are female and 89% have a college degree.
- The average respondent age is 36.8 years.
- Average annual base pay is \$86,640.
- The average firm is large, with about \$205 billion in assets.
- The average industry female high-skill share is 0.41.

Economic message: The baseline sample is tilted toward high-skilled workers and large public firms. This is appropriate for studying competition for skilled female talent, but it motivates later external-validity tests.

Table 2
Summary statistics

	Mean	Percentile			SD	N
		25th	50th	75th		
Review-level variables						
Maternity benefit rating	3.86	3	4	5	1.21	3,157
Current job	0.68	0	1	1	0.47	3,157
Female	0.46	0	0	1	0.50	3,157
College	0.89	1	1	1	0.31	3,157
Age	36.8	30	35	42	10.1	3,157
Base pay (USD)	86,640	53,100	87,530	119,460	38,536	2,994
Firm-level variables						
Assets (USD, mm)	204,995	4,395	24,238	110,450	501,883	3,157
Profit margin	0.09	0.04	0.09	0.16	0.12	3,157
Debt/Assets	0.23	0.11	0.20	0.32	0.17	3,157
q	3.23	1.40	2.17	3.66	2.85	3,157
Industry-level variables						
% female high-skill	0.41	0.30	0.38	0.52	0.13	3,157

This table displays summary statistics for the sample used in the baseline analysis. The unit of observation is at the maternity benefit review level. The sample reflects all employer maternity benefit reviews from July 2014 through April 2019 from respondents with nonmissing individual characteristics. We require that all respondents in the sample are regular employees of the firm, either currently or previously. We drop part-time, contractor, and other nonregular employees, for whom benefits may not be available. The sample is further restricted to reviews of firms that can be matched to Compustat. Appendix A.2 describes all variables.

overrepresents high-skilled workers. The fact that our sample is biased toward more educated, younger, and higher-paid workers does not seem, however, to affect our results.

5.3 Table 3: baseline industry female high-skill result

Read:

- The coefficient on % female high-skill is negative in all three specifications.
- Column 1 estimate: -0.654 .
- Column 2 estimate with state fixed effects: -0.641 .
- Column 3 estimate including base pay: -0.779 .

- Current-job reviews are rated higher, and firm size is positively related to ratings.
- Debt/assets is negatively related to maternity benefit quality in some specifications.

Economic message: Firms in industries with fewer high-skilled women offer better maternity benefits. This is the paper’s core sorting result.

Table 3
Maternity benefit ratings and industry female high-skill share

	Maternity benefit rating		
	(1)	(2)	(3)
% female high-skill	-.654** [.315]	-.641** [.312]	-.779** [.308]
ln(Age)	-.003 [.105]	-.037 [.100]	-.148 [.103]
Current job	.229*** [.050]	.217*** [.051]	.220*** [.056]
Female	-.058 [.043]	-.052 [.038]	-.052 [.048]
College	.147* [.078]	.149** [.073]	.110 [.088]
ln(Assets)	.084*** [.019]	.084*** [.018]	.087*** [.020]
Profit margin	-.260 [.409]	-.231 [.390]	-.123 [.461]
Debt/Assets	-.463** [.220]	-.401* [.205]	-.366 [.237]
q	.033*** [.012]	.030** [.012]	.030*** [.011]
ln(Base pay)			.066 [.053]
Year FE	Yes	Yes	Yes
State FE	No	Yes	Yes
N	3,157	3,157	2,594
R^2	.05	.07	.08

The dependent variable is the Glassdoor maternity benefit rating. The unit of observation is at the maternity benefit review level. % female high-skill is computed at the (four-digit NAICS) industry level. The sample includes all employer maternity benefit reviews from July 2014 through April 2019 from respondents with nonmissing individual characteristics. We require that all respondents in the sample are regular employees of the firm, either currently or previously. We drop part-time, contractor, and other nonregular employees, for whom benefits may not be available. The sample consists of reviews of firms that can be matched to Compustat. Appendix A.2 describes all variables. State fixed effects correspond to the respondent’s state. Standard errors are clustered at the industry level. *** $p < .01$; ** $p < .05$; * $p < .10$.

5.4 Table 4: robustness tests

Read:

- The negative coefficient remains after controlling for female executive share and gender pay gap.
- The result holds among female respondents only.
- The result holds among past-employee reviews only, reducing concerns about current-employee review management.
- The result survives dropping tech and finance industries.
- Inverse weighting by firm review count and imputing zero ratings for firms without maternity benefits also preserve the result.

Economic message: The baseline result is unlikely to be driven by respondent gender, firm review manipulation, overrepresented industries, review intensity, or omitted firm gender policies.

Table 4
Robustness tests

	Maternity benefit rating					
	Control for female executives	Control for gender pay gap	Female only	Past employees only	Drop tech & Finance	Inverse weighting
	(1)	(2)	(3)	(4)	(5)	(6)
% female high-skill	-.671** [.322]	-.768** [.319]	-.964** [.410]	-1.843*** [.340]	-1.313*** [.328]	-.761*** [.274]
ln(Age)	.019 [.120]	-.141 [.107]	.065 [.157]	-.002 [.183]	-.145 [.141]	-.222 [.155]
Current job	.195*** [.067]	.195*** [.059]	.201*** [.070]		.262*** [.073]	.228*** [.082]
Female	.005 [.009]	-.017 [.009]		.044 [.082]	-.110* [.061]	-.214** [.088]
College	.014 [.082]	.094 [.081]	.318*** [.149]	.004 [.124]	.203* [.120]	.238* [.127]
ln(Assets)	.120*** [.020]	.083*** [.020]	.118*** [.030]	.089*** [.024]	.019 [.018]	.070*** [.020]
Profit margin	.417 [.561]	-.090 [.468]	-.486 [.423]	.435 [.529]	-.188 [.412]	-.242 [.396]
Debt/Assets	-.362 [.276]	-.430 [.278]	-.420* [.248]	-.650*** [.281]	-.316* [.265]	-.312* [.182]
q	.037*** [.013]	.030*** [.010]	.041*** [.013]	.011 [.017]	.029** [.012]	.038** [.015]
ln(Base pay)	.062 [.062]	.079 [.056]	.077 [.081]	.173** [.076]	.129** [.064]	.057 [.077]
% female executives	-.306 [.224]					
Gender pay gap		.122 [.124]				
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
State FE	No	No	No	No	No	No
N	1,831	2,240	1,048	7,382	1,307	2,394
R^2	.11	.08	.12	.13	.12	.10

The dependent variable is the Glassdoor maternity benefit rating. The unit of observation is at the maternity benefit review level. % female high-skill is computed at the (four-digit NAICS) industry level. The sample includes all employer maternity benefit reviews from July 2014 through April 2019 from respondents with nonmissing individual characteristics. We require that all respondents in the sample are regular employees of the firm, either currently or previously. We drop part-time, contractor, and other nonregular employees, for whom benefits may not be available. The sample consists of reviews of firms that can be matched to Compustat. Column 1 controls for female executive share. Column 2 controls for firm gender pay gap. Column 3 keeps only female respondents. Column 4 drops all individuals reviewing their current job. Column 5 drops individuals working in the tech (NAICS codes: 3254, 3341, 3342, 3344, 3345, 3364, 5112, 5161, 5179, 5181, 5182, 5413, 5415, and 5417) or finance (NAICS codes between 5210 and 5259) industries. Column 6 weights observations by the inverse of the firm's Glassdoor observation count. Column 7 assumes that *Maternity benefit rating* equals zero for individuals at firms without maternity benefit ratings who also claim that their firm does not offer maternity benefits. Appendix A.2 describes all variables. State fixed effects correspond to the respondent's state. Standard errors are clustered at the industry level. *** $p < .01$; ** $p < .05$; * $p < .10$.

5.5 Table 5: female-friendly versus gender-neutral benefits

Read:

- Panel A shows that female-friendly benefit ratings are negatively related to industry female high-skill share.
- The coefficient is about -0.54 to -0.59 for female-friendly benefits.
- The analogous coefficients for gender-neutral benefits are smaller and statistically weaker.
- Panel B pooled regressions confirm that the relation is stronger for female-friendly benefits.

Economic message: Firms are not simply offering more benefits in general. They offer benefits that are especially attractive to women when female talent is scarce.

Table 5
Female-friendly versus gender-neutral benefit ratings and industry female high-skill share

A. Sample splits					
	Female-friendly benefit rating			Gender-neutral benefit rating	
	(1)	(2)	(3)	(4)	(5)
% female high-skill	-.575*** [.178]	-.541*** [.175]	-.588*** [.180]	-.220 [.146]	-.224 [.143]
ln(Age)	-.084** [.042]	-.089** [.041]	-.136*** [.046]	-.117** [.049]	-.114** [.049]
Current job	.149*** [.019]	.148*** [.019]	.153*** [.020]	.181*** [.019]	.181*** [.020]
Female	.054*** [.015]	.055*** [.014]	.053*** [.017]	.102*** [.023]	.106*** [.022]
College	.070*** [.021]	.069*** [.020]	.043* [.024]	.029 [.022]	.029 [.022]
ln(Assets)	.059*** [.010]	.058*** [.009]	.056*** [.010]	.038*** [.012]	.038*** [.011]
Profit margin	.080 [.167]	.088 [.159]	.154 [.186]	.156 [.180]	.161 [.168]
Debt/Assets	-.327*** [.092]	-.319*** [.087]	-.295*** [.094]	-.317*** [.075]	-.314*** [.072]
q	.031*** [.006]	.030*** [.005]	.028*** [.006]	.013 [.009]	.014* [.008]
ln(Base pay)			.025 [.023]		.025 [.025]
Year FE	Yes	Yes	Yes	Yes	Yes
State FE	No	Yes	Yes	No	Yes
Benefit FE	Yes	Yes	Yes	Yes	Yes
N	28,833	28,833	22,603	36,212	36,212
R ²	.07	.08	.08	.14	.14

(Continued)

Table 5
(Continued)

B. Pooled regressions				
	Benefit rating			
	(1)	(2)	(3)	(4)
% female high-skill×Female-Friendly	-.372*** [.129]	-.382*** [.142]	-.383*** [.120]	-.306*** [.109]
% female high-skill	-.201 [.148]	-.232 [.161]		
ln(Age)	-.104** [.042]	-.126*** [.046]	-.122** [.048]	
Current job	.167*** [.017]	.171*** [.019]	.130*** [.023]	
Female	.083*** [.016]	.089*** [.018]	.061*** [.022]	
College	.046** [.018]	.015 [.022]	.003 [.028]	
ln(Assets)	.047*** [.009]	.045*** [.010]		
Profit margin	.132 [.146]	.226 [.158]		
Debt/Assets	-.319*** [.072]	-.297*** [.020]		
q		.022*** [.006]		
ln(Base pay)		.038* [.022]	.020 [.017]	
Year FE	Yes	Yes	No	No
State FE	Yes	Yes	Yes	No
Benefit FE	Yes	Yes	Yes	Yes
Firm×Year FE	No	No	Yes	No
User-Job×Firm×Year FE	No	No	No	Yes
N	65,045	51,257	51,257	51,257
R ²	.11	.11	.35	.70

In panel A, the dependent variable is the Glassdoor female-friendly benefit rating in columns 1–3 and the gender-neutral benefit rating in columns 4–6. In panel B, the dependent variable is the Glassdoor benefit rating. The unit of observation is at the benefit review level. Ratings are normalized by benefit, whereby each rating is divided by the standard deviation of ratings in its benefit category. % *female high-skill* is computed at the (four-digit NAICS) industry level. *Female-friendly* takes a value of one for a female-friendly benefit and zero for a gender-neutral benefit. The sample in panel A includes all employer female-friendly (columns 1–3) and gender-neutral (columns 4–6) benefit reviews from July 2014 through April 2019 from respondents with nonmissing individual characteristics of firms that can be matched to Compustat. The sample in panel B includes all employer benefit reviews from July 2014 through April 2019 from respondents with nonmissing individual characteristics of firms that can be matched to Compustat. We require that all respondents are regular employees of the firm, either currently or previously. We drop part-time, contractor, and other nonregular employees, for whom benefits may not be available. Appendix A.2 describes all variables. State fixed effects correspond to the respondent's state. *Female-friendly* in panel B is absorbed by the fixed effects. Standard errors are double clustered at the user and industry level. *** $p < .01$; ** $p < .05$; * $p < .10$.

Table 5, Panel A: sample splits

Table 5, Panel B: pooled regressions

5.6 Table 6: actual paid weeks of maternity leave

Read:

- The dependent variable is $\ln(1 + \text{PaidWeeks})$.
- The coefficient on % female high-skill is negative across specifications.
- This means firms in low female high-skill industries offer more paid weeks of maternity leave.
- The result is important because paid weeks are an objective benefit measure rather than an employee rating.

Economic message: The evidence is not just about perceptions. Scarce female talent is associated with actual, measurable maternity leave generosity.

Table 6
Weeks of paid maternity leave and industry female high-skill share

	ln(1+Paid weeks)			
	(1)	(2)	(3)	(4)
% female high-skill	-.776* [.421]	-.724* [.400]	-.808* [.414]	-.707* [.419]
ln(Assets)	.082*** [.031]	.085*** [.031]	.087*** [.031]	.095*** [.032]
Profit margin	-.388 [.363]	-.395 [.364]	-.422 [.356]	-.478 [.348]
Debt/Assets	-.485* [.274]	-.520* [.273]	-.538* [.277]	-.545* [.287]
q	.040*** [.015]	.037** [.014]	.033** [.014]	.037*** [.013]
ln(Med. respondent age)		-.361*** [.181]	-.369*** [.179]	-.682*** [.198]
% female respondents		-.072 [.085]	-.071 [.085]	-.103 [.098]
% college respondents		-.181 [.133]	-.221 [.143]	-.368** [.184]
ln(Med. respondent base pay)				.213* [.108]
Region FE	No	No	Yes	Yes
N	415	415	415	350
R ²	.07	.09	.11	.14

The dependent variable is one plus weeks of paid maternity leave (in logarithm) available from Fairygodboss. The unit of observation is at the firm level without time-series variation. % *female high-skill* is computed at the (four-digit NAICS) industry level. The sample consists of firms in our baseline sample that can be matched to Fairygodboss. % *female high-skill*, *ln(Assets)*, *Profit margin*, *Debt/Assets*, and *q* reflect the latest year available in our sample. Appendix A.2 describes all variables. Region fixed effects are used instead of state fixed effects, given the smaller sample size, and take the value one if a firm has employees in a given Census division (New England, Middle Atlantic, East North Central, West North Central, South Atlantic, East South Central, West South Central, Mountain, and Pacific). Standard errors are clustered at the industry level. *** $p < .01$; ** $p < .05$; * $p < .10$.

5.7 Table 7 and Table 8: state-level and IV evidence

Read:

- Table 7 repeats the baseline logic using state-level female high-skill share.
- The coefficient on state female high-skill share is negative and statistically significant.
- Table 8 uses WWII mobilization rates as an instrument for state female high-skill share.
- The first stage is significant; the reported first-stage F-statistic is 10.71.
- The IV second-stage coefficient is negative and significant, and larger in magnitude than OLS.

Economic message: The state-level and IV results support external validity and reduce concerns that industry results are driven by unobserved industry characteristics.

Table 7
Maternity benefit ratings and state female high-skill share

	Maternity benefit rating		
	(1)	(2)	(3)
% female high-skill (state)	-3.205** [1.277]	-2.435** [1.099]	-2.645** [1.180]
ln(Age)	.109* [.056]	.121** [.052]	.037 [.071]
Current job	.412*** [.025]	.391*** [.022]	.380*** [.028]
Female	-.191*** [.038]	-.113*** [.039]	-.111** [.049]
College	.111** [.048]	.101** [.048]	.052 [.054]
ln(Base pay)			.090*** [.027]
Year FE	Yes	Yes	Yes
Sector FE	No	Yes	Yes
N	10,385	9,318	6,445
R ²	.04	.07	.08

The dependent variable is the Glassdoor maternity benefit rating. The unit of observation is at the maternity benefit review level. The sample includes all employer maternity benefit reviews from July 2014 through April 2019 from respondents with nonmissing individual characteristics. We require that all respondents in the sample are regular employees of the firm, either currently or previously. We drop part-time, contractor, and other nonregular employees, for whom benefits may not be available. The sample consists of both public and private firms on Glassdoor. % female high-skill (state) is computed at the state level. Appendix A.2 describes all variables. Sector fixed effects reflect the respondent's sector and correspond to Glassdoor's 25 sector classification system. Standard errors are clustered at the state level. *** $p < .01$; ** $p < .05$; * $p < .10$.

Table 7: state female high-skill share

Table 8
Maternity benefit ratings and state female high-skill share

	Maternity benefit rating		
	OLS	IV	
		1st stage	2nd stage
	(1)	(2)	(3)
% female high-skill (state)	-6.850*** [2.081]		-11.020*** [4.002]
% white (1940)	3.321 [2.415]	-.419*** [.114]	1.961 [2.308]
% black (1940)	2.562 [1.995]	-.377*** [.096]	1.604 [1.815]
% male (1940)	2.350 [6.123]	1.245*** [.346]	7.643 [6.260]
ln(GDP per capita)	-.573 [.377]	-.031 [.027]	-.733** [.322]
Cook Partisan Lean Index	-.014 [.009]	-.001** [.000]	-.018** [.008]
WWII mobilization rate		-1.057*** [.329]	
1940 age buckets	Yes	Yes	Yes
N	48	48	48
R ²	.66	.88	.63
First-stage F-stat		10.71	

The dependent variable is the state-level average Glassdoor maternity benefit rating. The unit of observation is at the state level. The sample includes all employer maternity benefit reviews from July 2014 through April 2019 from respondents with nonmissing individual characteristics. We require that all respondents in the sample are regular employees of the firm, either currently or previously. We drop part-time, contractor, and other nonregular employees, for whom benefits may not be available. The sample consists of both public and private firms on Glassdoor. Observations are weighted by the number of Glassdoor maternity benefit reviews per state. Maternity benefit rating and % female high-skill (state) are collapsed at the state level by taking averages. In the IV specification in columns 2 and 3, % female high-skill (state) is instrumented using WWII mobilization rate. 1940 age buckets represent age bins that track the fraction of each state's 1940s population belonging in each 5-year bin (i.e., 0-4, 5-9, 10-14, ..., 70-74, 75+). Appendix A.2 describes all variables. Standard errors are robust. *** $p < .01$; ** $p < .05$; * $p < .10$.

Table 8: IV using WWII mobilization

5.8 Table 9: maternity benefits and female employment

Read:

- The dependent variable is the percentage of female employees at the firm.
- Median maternity benefit rating alone is not strongly positive.
- The interaction between median maternity benefit rating and low female high-skill industry is positive and significant.
- In the most robust specification, a one-point increase in maternity rating is associated with a 2.6 percentage point increase in female employment for firms in low female high-skill industries.

Economic message: Better maternity benefits are associated with more female employees precisely where female talent is scarce. This supports the proposed mechanism.

	% female				
	(1)	(2)	(3)	(4)	(5)
Median maternity benefit rating	−.826 [.514]	−.829 [.521]	−.889* [.527]	−1.068 [.667]	−.780 [.787]
Median maternity benefit rating ×Low % female high-skill	3.576*** [.887]	3.577*** [.896]	3.683*** [.901]	3.150*** [1.058]	2.594** [1.076]
Low % female high-skill	−35.634*** [3.493]	−35.655*** [3.447]	−35.973*** [3.533]	−27.584*** [4.408]	−25.385*** [4.477]
ln(Assets)	−.827 [.512]	−.834 [.523]	−1.013** [.494]	−.880* [.512]	−.763 [.557]
Profit margin	−9.153 [7.280]	−9.131 [7.352]	−9.923 [6.680]	−9.130* [4.606]	−8.843* [4.680]
Debt/Assets	−3.751 [5.480]	−3.717 [5.426]	−6.108 [5.772]	−9.861* [5.871]	−9.394 [6.556]
<i>q</i>	−.833** [.351]	−.832** [.355]	−.907*** [.338]	−.866** [.330]	−.835** [.332]
ln(Median respondent age)	.346 [2.560]	−.121 [2.445]	−.121 [2.681]	.468 [2.681]	.001 [3.091]
% college respondents	.105 [2.593]	.174 [2.542]	.599 [2.251]	.820 [2.503]	.001 [1.809]
ln(Median respondent base pay)					
Region FE	No	No	Yes	Yes	Yes
Sector FE	No	No	No	Yes	Yes
N	343	343	343	319	270
<i>R</i> ²	.54	.54	.56	.66	.66

The dependent variable is the percentage of female employees per firm. The unit of observation is at the firm level, and we do not observe a time series. The sample consists of our baseline sample's Glassdoor firms that report gender composition data in a Diversity and Inclusion report or a Diversity and Inclusion section within a Corporate Social Responsibility report. *Low % female high-skill* is an indicator equal to one if the firm operates in a below median % female high-skill industry, and zero otherwise. *ln(Assets)*, *Profit margin*, *Debt/Assets*, and *q* reflect the latest year available in our sample. Appendix A.2 describes all variables. Region fixed effects are a set of dummies that take the value one if a firm has employees in a given Census division (New England, Middle Atlantic, East North Central, West North Central, South Atlantic, East South Central, West South Central, Mountain, and Pacific). Sector fixed effects reflect the respondent's sector and correspond to Glassdoor's 25 sector classification system. Standard errors are clustered at the industry level. *** $p < .01$; ** $p < .05$; * $p < .10$.

5.9 Table 10: event study around paid family leave laws

Read:

- The dependent variable is the 3-day abnormal return around state paid family leave law passage.
- The key interaction is *Law* × *HighMaternity* × *LowFemaleHighSkill*.
- The triple interaction is negative and statistically significant across industry-based and state-based definitions.
- Industry-based estimates imply abnormal returns of roughly −1.0 to −1.4%.
- State-based estimates imply abnormal returns of roughly −1.2 to −2.0%.

Economic message: Voluntary maternity benefits created value for firms in scarce female-talent markets. When state mandates reduced the relative advantage of those benefits, the affected firms lost value.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
	CAR [-1,1]									
Low × High mat. × Low % firm high-skill (ind)	−.011** [.003]	−.011* [.006]	−.010** [.005]	−.011** [.004]	−.012*** [.001]					
Low × Low % firm high-skill (ind)	.007** [.004]	.007 [.004]	.008 [.003]	.009* [.005]	.012** [.001]					
High mat. × Low % firm high-skill (ind)	.001 [.001]	−.000 [.001]	.000 [.001]	.001 [.001]	.000 [.001]					
Low × High maternity	.002 [.001]	.002 [.001]	.004 [.001]	.001 [.001]	.000*** [.001]	.008*** [.001]	.000** [.002]	.001*** [.002]	.003* [.001]	.003* [.001]
High maternity	.000 [.001]	−.000 [.001]	−.000 [.001]	.001 [.004]	−.001 [.001]	−.001 [.001]	.002 [.001]	.002 [.001]	.001 [.001]	.001 [.001]
Low × High mat. × Low % firm high-skill (state)	.000 [.002]	.000 [.001]	.000 [.001]	.000 [.001]	−.002*** [.001]	−.002*** [.001]	−.011** [.001]	−.011** [.001]	−.012* [.001]	−.012* [.001]
High mat. × Low % firm high-skill (state)	.000 [.002]	.000 [.001]	.000 [.001]	.000 [.001]	−.002 [.002]	−.002 [.001]	−.002 [.001]	−.002 [.001]	−.002 [.001]	−.002 [.001]
ln(Assets)	.001*** [.001]	.001** [.001]	.001 [.001]	.002 [.001]	.001 [.001]	.001*** [.001]	.001** [.001]	.003 [.001]	.001 [.001]	.001 [.001]
Profit margin	.000 [.001]	.000 [.001]	.000 [.001]	.000 [.001]	.000 [.001]	.000 [.001]	.000 [.001]	.000 [.001]	.000 [.001]	.000 [.001]
Debt/Assets	.001 [.001]	.001 [.001]	.001 [.001]	.001 [.001]	.001 [.001]	.001 [.001]	.001 [.001]	.001 [.001]	.001 [.001]	.001 [.001]
<i>q</i>	.001 [.001]	.001 [.001]	.001 [.001]	.001 [.001]	.001 [.001]	.001 [.001]	.001 [.001]	.001 [.001]	.001 [.001]	.001 [.001]
Industry × Event FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
State × Event FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Industry × State FE	No	No	No	No	No	No	No	No	No	No
Firm FE	No	No	No	Yes	Yes	No	No	No	Yes	Yes
N	3,147	3,135	3,133	3,135	3,135	3,135	3,144	3,144	3,144	3,144
<i>R</i> ²	.50	.50	.55	.60	.57	.49	.49	.54	.59	.56

The dependent variable is the cumulative abnormal return over a [−1, +1] window (CAR) around the signing date of legislation either enacting or expanding state-sponsored paid family leave programs. We consider treatments or exposures within our sample period (DC, CA, HI, NY, MA, NH, NJ, CT, and OR). CARs are calculated using a CAPM market model in columns 1–4 and 5–9, and a Carhart four-factor model in columns 7 and 10 (Carhart 1997). The unit of observation is at the firm-event level. The sample includes all firms with employer maternity benefit reviews from July 2014 through April 2019 from corporations with non-zero industry observations of firms that can be matched to Compustat and CRSP. We require that all respondents in the sample not be regular employees of the firm, either currently or previously. We drop past state, contracts, and other nonlegislative exposures, for which results may not be available. *Low* is an indicator variable equal to one for firms headquartered in states with paid family leave programs at the time of enactment, and zero otherwise. *Low % female high-skill (ind)* is an indicator variable equal to one if a firm has above median industry % female high-skill personnel. *Low % female high-skill (state)* is an indicator variable equal to one if a firm has above (below) median % female high-skill (state) personnel. *High maternity* is an indicator variable equal to one (zero) if a firm's average percentage maternity benefit rating is above (below) the industry median in columns 1–5, and the state median in columns 6–10. Appendix A.2 describes all variables. Industry × Event fixed effects correspond to the firm's four-digit NAICS industry interacted with each law passage event. State × Event fixed effects correspond to the firm's headquarters state interacted with each law passage event. Industry × State fixed effects correspond to the firm's four-digit NAICS industry interacted with its headquarters state. Standard errors are clustered at the state level. *** $p < .01$; ** $p < .05$; * $p < .10$.

5.10 Appendix tables: sample construction and variable definitions

Read:

- Table A.1 documents how the raw Glassdoor maternity review sample is reduced to the public and public+private samples.
- Table A.2 defines key review-level, firm-level, industry-level, and state-level variables.

Economic message: These appendix tables clarify that the empirical design relies on a carefully filtered sample and on multiple externally linked data sources.

Table A.1
Sample construction

	Sample size	
All Glassdoor (GD) maternity benefit reviews (Raw Data)	45,217	
Drop part-time, contract, and other nonregular employees	-4,521	
GD maternity benefits sample with only regular employees	40,696	
Drop if missing data on age, sex, education, or current job status	-30,311	
GD maternity benefits sample with only regular employees and nonmissing respondent data	10,385	Public+private sample
Drop nonpublic firms	-7,228	
GD maternity benefits sample with only reg. emps., nonmissing reviewer data, and public firms	3,157	Public sample

This table describes the steps we follow to construct the sample of maternity benefit reviews of public and of public+private firms used in our regression analyses from the raw Glassdoor reviews data. Each observation in the raw Glassdoor data reflects an individual's review of a maternity benefit for a given firm (i.e., User 987 assigned a 3/5 rating for Maternity Benefits at ExampleCo).

Table A.1: sample construction

Table A.2
Variable descriptions

Review-level variables	
<i>Maternity benefit rating</i>	Glassdoor respondent assigned rating ranging from 1 to 5 for the employer's maternity leave benefits. <i>Source:</i> Glassdoor
<i>Female-friendly benefit rating</i>	Glassdoor respondent assigned rating ranging from 1 to 5, where the following benefits are considered female-friendly benefits: Adoption Assistance, Childcare, Dependent Care, Diversity Program, Family Medical Leave, Fertility Assistance, Flexible Savings Account, Health Insurance, Health Savings Account, Mental Health Care, Maternity Leave, Reduced or Flexible Hours, Vacation & Paid Time Off, and Work from Home. <i>Source:</i> Glassdoor
<i>Gender-neutral benefit rating</i>	Glassdoor respondent assigned rating ranging from 1 to 5 for all benefits that are considered gender-neutral benefits. The specific benefits are: Accidental Death & Dismemberment Insurance, Apprenticeship Program, Bereavement Leave, Charitable Gift Matching, Commuter Checks & Assistance, Company Car, Company Social Events, Dental Insurance, Disability Insurance, Employee Assistance Program, Employee Discount, Employee Stock Purchase Plan, Equity Incentive Plan, Free Lunch or Snacks, Gym Membership, Health Care On-Site, Job Training, Legal Assistance, Life Insurance, Military Leave, Mobile Phone Discount, Occupational Accident Insurance, Paid Holidays, Pension Plan, Pet Friendly Workplace, Professional Development, Retiree Health & Medical, Retirement Plan, Sabbatical, Sick Days, Stock Options, Supplemental Life Insurance, Supplemental Workers' Compensation, Travel Concierge, Tuition Assistance, Unpaid Extended Leave, Vision Insurance, and Volunteer Time Off. <i>Source:</i> Glassdoor
<i>Female-friendly</i>	Indicator variable equal to one (zero) if a benefit is a female-friendly (gender-neutral) benefit <i>Source:</i> Glassdoor
<i>Current job</i>	An indicator variable equal to one (zero) if the respondent reviews his/her current (prior) job <i>Source:</i> Glassdoor
<i>Female</i>	An indicator variable equal to one (zero) if a respondent is female (male) <i>Source:</i> Glassdoor
<i>ln(Age)</i>	Natural logarithm of Glassdoor respondent's age, calculated as the response year minus the respondent's birth year, winsorized at the 1% level <i>Source:</i> Glassdoor
<i>College</i>	An indicator variable equal to one (zero) if a respondent has (does not have) a college degree <i>Source:</i> Glassdoor
<i>ln(Base pay)</i>	Natural logarithm of Glassdoor respondent's annual base pay, winsorized at the 1% level <i>Source:</i> Glassdoor
Firm-level variables	
<i>Median maternity benefit rating</i>	Firm-level median maternity benefit rating <i>Source:</i> Glassdoor
<i>ln(Assets)</i>	Natural logarithm of firm's total assets, winsorized at the 1% level <i>Source:</i> Compustat
<i>Profit margin</i>	Net income divided by total revenue, winsorized at the 1% level <i>Source:</i> Compustat
<i>Debt/Assets</i>	Total debt divided by total assets, winsorized at the 1% level <i>Source:</i> Compustat
<i>Q</i>	The sum of market capitalization and total debt, divided by the sum of total book equity and total debt, winsorized at the 1% level <i>Source:</i> Compustat
<i>% female executives</i>	% of total firm executives that are female, where individuals identified in Execucomp are considered executives <i>Source:</i> Execucomp
<i>Gender pay gap</i>	Ratio of firm average female base pay to male base pay <i>Source:</i> Glassdoor
<i>% female respondents</i>	Firm-level share of respondents who are female <i>Source:</i> Glassdoor
<i>ln(Median respondent age)</i>	Firm-level median respondent age (log-transformed), winsorized at the 1% level <i>Source:</i> Glassdoor
<i>% college respondents</i>	Firm-level share of respondents who have a college degree <i>Source:</i> Glassdoor
<i>ln(Median respondent base pay)</i>	Firm-level median base respondent base pay, winsorized at the 1% level (log-transformed) <i>Source:</i> Glassdoor
<i>Paid weeks</i>	Paid weeks of maternity leave offered, winsorized at the 1% level <i>Source:</i> Fairygodboss
<i>% female</i>	Firm-level share of female employees, winsorized at the 1% level <i>Source:</i> Corporate Disclosures

(Continued)

Table A.2: variable definitions, part 1

Table A.2 (Continued)	
Industry-level variables	
<i>% female high-skill</i>	Total female college workers divided by total college workers in the respondent's firm's industry-year cell, where industry is defined at the four-digit NAICS level <i>Source:</i> QWL
State-level variables	
<i>% female high-skill (state)</i>	Total female college workers divided by total college workers in the respondent's state-year <i>Source:</i> QWL
<i>WWII mobilization rate</i>	Estimated fraction of men in a given state who served in the U.S. Armed Forces during WWII, winsorized at the 1% level <i>Source:</i> Selective Service System Monographs
<i>% white (1940)</i>	State share of U.S. population in 1940 that is white <i>Source:</i> Census
<i>% black (1940)</i>	State share of U.S. population in 1940 that is black <i>Source:</i> Census
<i>% male (1940)</i>	State share of U.S. population in 1940 that is male <i>Source:</i> Census
<i>ln(GDP per capita)</i>	Natural logarithm of the state's average GDP per capita over the sample period <i>Source:</i> Bureau of Economic Analysis
<i>Cook Partisan Lean Index</i>	Measure of a state's political leaning relative to the nation as a whole. A value of zero indicates that the state's Republican-Democrat lean is at the national average. Higher index values reflect more Republican leaning states. The index reflects 2016 values, the latest available from the Cook Political Report within the sample period <i>Source:</i> Cook Political Report
Legislative-event-level variables	
<i>CAR[-1,+1]</i>	Three-day cumulative abnormal returns (CARs) around the passage of state paid family leave laws, winsorized at the 1% level. CARs are calculated under either a CAPM market model or a Carhart four-factor model (Carhart 1997) <i>Source:</i> CDSIF
<i>Law</i>	Indicator variable that takes the value one for observations in states that passed a paid family leave law at the time of law passage, and zero otherwise <i>Source:</i> State Governments

6 Short summary

- The paper asks why firms provide costly nonwage benefits rather than equivalent cash pay.

- It proposes a sorting mechanism: benefits that are more valuable to a target worker group help firms attract that group.
- The empirical focus is maternity leave benefits, which are disproportionately valuable to women.
- Firms in industries with fewer high-skilled women offer better maternity benefits.
- The result is robust to respondent controls, firm controls, female-only samples, past-employee samples, dropping tech and finance, inverse weighting, and zero-rating imputations.
- The result generalizes to female-friendly benefits but not to gender-neutral benefits.
- Actual paid weeks of maternity leave show the same pattern.
- State-level evidence and IV evidence using WWII mobilization rates support the labor-scarcity interpretation.
- Firms with stronger maternity benefits have higher female employment in industries where female talent is scarce.
- Event-study evidence shows that state paid leave mandates reduce the value of voluntary maternity benefits for high-maternity firms in scarce female-talent markets.

7 Conclusion

7.1 Core conclusion

Liu, Makridis, Ouimet, and Simintzi show that firms use nonwage benefits strategically. Maternity benefits are more generous when high-skilled female labor is scarce, suggesting that firms use female-friendly benefits to compete for women and increase gender diversity.

7.2 Economic intuition

Cash wages are broadly valued by all employees, but maternity benefits are differentially valuable to women and can signal a female-friendly workplace. When women are scarce in the relevant labor market, the marginal value of attracting and retaining them is higher. Firms therefore find it efficient to offer better maternity benefits.

7.3 Takeaway

Nonwage benefits are part of firms' labor-market strategy. They can be used to target specific worker groups, and they can create firm value when they help firms attract scarce talent. The distribution of benefits across firms is therefore shaped by labor-market scarcity, not only by firm size, profitability, or general generosity.